

This examination is part of the assessment of students' abilities as follows:

		Question
CLO 1	Explain and use the theory of sets and its applications.	Q1-Q3
CLO 2	Understand and analyze mathematical contents related to real number system and its properties and solving polynomial equations and inequalities.	Q4-Q5
CLO 3	Explain and enable to use some concepts of cartesian product, relation, inverse of a relation, and domain and range of a relation.	Q6-Q8
CLO 4	Have moral and ethical values, respect rights, listen to others' opinions, and able to work collaboratively with others.	-

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Question 1.

1.1 Given $A = \{-3, -2, 1, 5\}$. If $A \cup B = \{-3, -2, -1, 1, 4, 5, 6\}$ and $A \cap B = \{-3, 1\}$. Find the set B .

(3 points)

1.2 Given $A = \{1, 2, 3, 4\}$ and $B = \{1, 3, 4, 5, 6\}$. If $C \subset A$, $C \subset B$, $4 \notin C$, and $n(C) > 1$. Find $P(C)$.

(4 points)

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Question 2. Let U be the universal set, and let A and B be subsets of U such that $A \subset B$.

Use the above information to answer the questions 2.1 and 2.2.

2.1 Use a Venn-Euler diagram to shade the region representing $(A' \cap B') \cup (A \cap B)$. (3 points)

2.2 Find an equivalent expression for $(A' \cap B') \cup (A \cap B)$ using complement and set difference notations. (3 points)

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Question 3.

3.1 Given $A = \{x \mid 5 \leq x < 10\}$, $B = \{x \mid 2 < x \leq 6\}$, and $C = \{x \mid 3 < x < 7\}$. Find the set $A - (B \cup C)$.
(3 points)

3.2 Let U be the universal set, and let A and B be subsets of U . Suppose that $n(U) = 100$, $n(A \cap B) = 35$, and $n(A' \cap B') = 9$. If $n(A) \geq 61$, what is the maximum possible value of $n(B)$?
(6 points)

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Question 4.

4.1 Convert the repeated decimal $0.8\overline{324}$ to a fraction.

(2 points)

4.2 Determine the interval of x^2 when $-8 < x < 6$.

(2 points)

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4.3 Given $p(x) = x^3 + (k-1)x^2 - k^3$, where k is a negative real number. If the remainder when $p(x)$ is divided by $x-3$ is 18, then what is the remainder when $p(x)$ is divided by $2x+1$? (5 points)

4.4 Consider the operation $a*b = a+b+ab$ on real numbers.

Find the inverse of 3 under this operation.

(5 points)

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Question 5.

5.1 Solve the equation $2x^3 + x^2 - 12x + 9 = 0$ using the synthetic division.

(3 points)

5.2 Find the solution of the inequality $(x+2)^2(x+1)(1-x) \geq 0$.

(4 points)

5.3 Find the solution of the inequality $\frac{2}{x+4} \leq \frac{1}{x-2}$.

(4 points)

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Question 6. Let A and B are sets such that $A \times B = \{(1,1), (1,x), (1,y), (2,1), (2,x), (2,y)\}$.

(a) Find the sets A and B . (2 points)

(b) Find $n(A \cap B)$. (1 points)

(c) Find the total number of relations from B to A . (2 points)

(d) Let r be a relation from B to A such that $n(r) = 4$ and $1 \notin D_r$.

Give an example of the relation r . (2 points)

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Question 8.

8.1 Given the following relations $r_1 = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y \leq x\}$ and $r_2 = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y \geq x^2\}$.

Let $r = r_1 \cap r_2$. Sketch the graph of r and find D_r and R_r .

(5 points)

8.2 Given the relation $r = \left\{ (x, y) \in \mathbb{R} \times \mathbb{R} \mid y = \frac{x+5}{3x-6} \right\}$. Find r^{-1} and $R_{r^{-1}} - D_{r^{-1}}$.

(6 points)

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